Diagnostics: Report

Findings from the Diagnostics class and its unit tests.

Summary

The Diagnostics class holds the checks that both of our solvers share. It checks the inputs before a solve starts, decides when to stop, spots a derivative that is too small, and keeps the iteration history. In our experiments every bad input was rejected with a clear message, the stopping rule did what we expected, and all 10 unit tests passed.

1. Input Validation

When an input is wrong, the class raises a `ValueError` that names the problem and shows the value. Good inputs go through. For $f(x) = x^2 - 2$, the interval [0, 2] is accepted and gives $f(a) = -2$ and $f(b) = 2$. An interval where one end is already a root, like [0, 1] for $f(x) = x$, is also accepted, because $f(a)f(b) = 0$ is not a sign-change failure.

Input	Message reported
tol = -1	Tolerance must be a positive number, got -1
tol = NaN	Tolerance must be a positive number, got nan
max_iter = 0	max_iter must be a positive integer, got 0
max_iter = 2.5	max_iter must be a positive integer, got 2.5
f = 5 (not a function)	f must be a callable function
Interval [3, 1]	Interval must satisfy a < b, got a=3, b=1
Interval [2, 3]	No sign change on [2, 3]: f(a)=2, f(b)=7
Interval [0, inf]	b must be a finite number, got inf

2. Stopping Rule

A run stops as converged when the error (the step size for Newton, half the interval width for bisection) or the residual $|f(x)|$ is below `tol`. It stops as failed when `max_iter` is reached. The comparison is strict, so a value exactly equal to `tol` does not count as converged.

To see this working, we wrote a small loop that calls Diagnostics on $f(x) = x^2 - 2$. It is only for trying out the class and is not one of our real solvers. For bisection on [0, 2], the test that stopped the run changed with the tolerance:

tol	Iterations	Stopped by	Final residual
1e-2	8	error and residual	4.27e-4
1e-4	14	residual only	8.20e-5
1e-6	21	error only	1.62e-6
1e-8	28	error and residual	5.24e-9
1e-10	30	residual only	3.15e-11

At tol = 1e-6 the half-width test stopped the run even though the residual (1.62e-6) was still above tol. At 1e-4 and 1e-10 the residual test fired first. This is why we keep both tests: with only one of them, some of these runs would have gone on for extra iterations.

Newton from x0 = 1 took 2, 3, 4, 4 and 4 iterations for tolerances from 1e-2 down to 1e-10. It always stopped on the residual, and once tol was 1e-6 or smaller it ended at a residual of about 4.5e-12.

3. Failure Detection

The derivative check catches a Newton step that would divide by a number very close to zero. The default cut-off is 1e-12 and it is strict, so a derivative of exactly 1e-12 is not flagged.

|f'(x)|	0	1e-15	1e-12	1e-6	2.0
Flagged as too small	Yes	Yes	No	No	No

We also reproduced two failures with the loop. On $x^2 - 1$ from x0 = 0, f'(0) = 0, so the run stopped before any division, after 0 iterations, with the message "zero derivative". On bisection with tol = 1e-14 and `max_iter = 5`, the run ended with converged = False after 5 iterations.

4. Unit Tests

The tests are in `tests/test_diagnostics.py` and use `unittest`. Before every test, `setUp` creates a fresh Diagnostics object (tol = 1e-6, max_iter = 50) and the function $x^2 - 2$. Running `python -m unittest tests.test_diagnostics` gives 10 tests, all passing.

Group	Tests	What is checked
Input validation	5	Negative tolerance, max_iter = 0, valid interval gives (−2, 2), no sign change on [2, 3], a not less than b
Failure detection	1	Derivative 0 is flagged, derivative 2 is not
Stopping rule	3	Small error converges, small residual converges, both large does not converge
History and result	1	After one record, the result shows 1 iteration, residual 0.25 and converged = True

5. Limitations

The short test file does not cover NaN or infinite inputs, the copy of the history that `result` returns, or `print_table`. The class handles these, but they are not tested there. Python treats True as the integer 1, so `Diagnostics(tol=True)` is accepted as a tolerance of 1. The derivative cut-off stays at 1e-12 unless a different `eps` is passed in.

Conclusion

Diagnostics gives both solvers one common way to reject bad input, decide when to stop, and report a failure. Using the error and the residual together makes the stopping rule more reliable, and the unit tests show that each check behaves as we intended. Next, we could add tests for the cases listed in the limitations.